

Grupo I

1) a - F

b - V

$$c - \begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & -2 \\ 3 & -2 & -3 \end{bmatrix} \xrightarrow[\substack{L_2 \leftarrow L_2 - 2L_1 \\ L_3 \leftarrow L_3 - 3L_1}]{OL_1} \begin{bmatrix} 1 & 1 & -1 \\ 0 & -5 & 0 \\ 0 & -5 & 0 \end{bmatrix} \xrightarrow[\substack{L_3 \leftarrow L_3 - L_2}]{OL_2} \begin{bmatrix} 1 & 1 & -1 \\ 0 & -5 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

(V)

d - V

2) a - V

b - $\dim(\mathcal{N}(A)) = n - \text{Con}(A)$

$$A = \begin{pmatrix} 1 & -1 & 1 & 0 \\ -1 & 0 & 1 & -1 \\ 1 & 1 & 0 & 2 \\ 1 & -1 & -1 & 0 \end{pmatrix} \xrightarrow[\substack{L_2 \leftarrow L_2 + L_1 \\ L_3 \leftarrow L_3 - L_1 \\ L_4 \leftarrow L_4 - L_1}]{OL_1} \begin{pmatrix} 1 & -1 & 1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 2 & -1 & 2 \\ 0 & 0 & -2 & 0 \end{pmatrix}$$

$$\xrightarrow[\substack{L_3 \leftarrow L_3 + 2L_2}]{OL_2} \begin{pmatrix} 1 & -1 & 1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & -2 & 0 \end{pmatrix} \xrightarrow[\substack{L_4 \leftarrow L_4 + \frac{2}{3}L_3}]{OL_3} \begin{pmatrix} 1 & -1 & 1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\text{Con}(A) = 3$$

$$\dim(\mathcal{N}(A)) = 4 - 3 = 1$$

(F)

$$c - \begin{pmatrix} 1 & -1 & 1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{L_3 \leftarrow \frac{1}{3}L_3} \begin{pmatrix} 1 & -1 & 1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

(V)

d - F

$$3) \quad a - \begin{pmatrix} 1 & 1 & 2 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 5 \\ 5 \\ 1 \end{pmatrix} \quad v$$

$$b - \dim(\text{Im } \phi_0) = 2$$

Ⓕ

$$c - \phi_k \begin{pmatrix} k & 1 & 2 & 0 \\ 0 & k+1 & 1 & 1 \\ 0 & 0 & k & -1 \end{pmatrix}$$

$$\dim(\text{Im}) \leq 3 \quad \leftarrow \begin{matrix} n^\circ \text{ linhas} \\ \swarrow \end{matrix}$$

n° columns $\rightarrow$

$$m = \dim(\text{Nuc}) + \dim(\text{Im})$$

$$4 = \dim(\text{Nuc}) + \dim(\text{Im})$$

$$4 \geq \dim(\text{Nuc}) \geq 1$$

Logo nunca é igual a 0

Ⓖ

d -

II

$$1) a- \quad A - \lambda I = \begin{pmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{pmatrix}$$

$$\det(A - \lambda I) = -\lambda^3 + 1 + 1 + \lambda + \lambda + \lambda \\ = -\lambda^3 + 3\lambda + 2$$

$$-\lambda^3 + 3\lambda + 2 = 0$$

$$\Leftrightarrow \lambda^3 - 3\lambda - 2 = 0$$

$$\text{Para } \lambda = 2$$

$$\begin{array}{c|cccc} & 1 & 0 & -3 & -2 \\ 2 & & 2 & 4 & 2 \\ \hline & 1 & 2 & 1 & 0 \end{array}$$

$$(x-2)(x^2+2x+1) = 0$$

$$(x-2)=0 \quad \vee \quad x^2+2x+1=0$$

$$\Leftrightarrow x=2 \quad \vee \quad x=-1$$

$$b- \text{ Enunciamos o núcleo de } \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \xrightarrow{O2} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$x_1 + x_2 + x_3 = 0$$

$$\Leftrightarrow x_1 = -x_2 - x_3$$

$$v = (-x_2 - x_3, x_2, x_3)$$

$$v = (-x_2, x_2, 0) + (-x_3, 0, x_3)$$

$$= x_2(-1, 1, 0) + x_3(-1, 0, 1)$$

$$= \langle (-1, 1, 0), (-1, 0, 1) \rangle$$

C- Para o valor próprio -1

$$(-1, 1, 0), (-1, 0, 1)$$

Para o valor próprio 2

$$\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \xrightarrow[\substack{L_2 \leftarrow L_2 + \frac{1}{2}L_1 \\ L_3 \leftarrow L_3 + \frac{1}{2}L_1}]{OL} \begin{pmatrix} -2 & 1 & 1 \\ 0 & -3/2 & 3/2 \\ 0 & 3/2 & -3/2 \end{pmatrix}$$